PROBLEM SET 2 DUE 3.22

1. Let $f : M \to N$ and $g : N \to M$ be A-module homomorphisms such that $gf = \mathrm{id}_M$. Show that $N = \mathrm{Im}\, f \oplus \ker g$.

2. Let M be an A-module, $N \subseteq M$ a submodule, $i : N \to M$ the inclusion homomorphism, and $p : M \to M/N$ the quotient homomorphism. Prove that the following are equivalent:

(i) There exists a submodule N' of M such that $M = N \oplus N'$.
(ii) There exists an A-module homomorphism $p : M \to N$ such that $pi = \mathrm{id}_N$.
(iii) There exists an A-module homomorphism $j : M/N \to M$ such that $qj = \mathrm{id}_{M/N}$. [1]

3. A submodule $N \subseteq M$ is called a **direct summand** if it satisfies the conditions in Problem **2**.

(1) Show that if N is a direct summand of M and L is a submodule of M containing N, then N is a direct summand of L.
(2) Give an example of a submodule which is not a direct summand.
(3) Show that if N is a direct summand of M, then $M \cong N \oplus (M/N)$. Find an example to show that the converse fails.

4. Show that $\mathbb{Q}$, considered as a $\mathbb{Z}$-module, is neither finitely generated nor free.

5. Let A be a nonzero ring, $\mathfrak{a} \subseteq A$ an ideal. Prove the following statements:

(1) Any two elements a, b in A are linearly dependent over A.
(2) The ideal $\mathfrak{a}$ is a free A-module iff $\mathfrak{a} = 0$ or $\mathfrak{a}$ is generated by a non-zero-divisor.
(3) $A/\mathfrak{a}$ is a free A-module iff $\mathfrak{a} = 0$ or $\mathfrak{a} = A$.
(4) If every A-module is free, then A is a field.

6. Let A be a nonzero ring and let $m, n \geqslant 1$. Prove the following statements:

(1) If there is an injective A-linear map $A^n \to A^m$, then $m \geqslant n$.

Hint: It suffices to show that if $\Phi \in M_{m\times n}(A)$ with $m < n$, then there exists a nonzero $\mathbf{x} \in A^n$ such that $\Phi\mathbf{x} = 0$. Suppose $\mathrm{rk}(\Phi) = r(< n)$. Assume without loss of generality that the top left r-by-r minor of Φ is nonzero. Construct $\mathbf{x}$ as follows: the first $r+1$ coordinates of $\mathbf{x}$ are the r-by-r minors, with appropriate signs, of the top left r-by-$(r+1)$ submatrix; the rest of the coordinates are all zero.

(2) If there is a surjective A-linear map $A^n \to A^m$, then $m \leqslant n$.

Hint: Show that if $\varphi : A^n \to A^m$ is surjective, then there exists $\psi : A^m \to A^n$ such that $\varphi\psi = \mathrm{id}$. An alternative strategy for solving (2) is reduction modulo $\mathfrak{m}$ where $\mathfrak{m}$ is any maximal ideal of A. Show that if $\varphi : A^n \to A^m$ is surjective, so is $\bar{\varphi} : A^n/\mathfrak{m}A^n \to A^m/\mathfrak{m}A^m$.

7. Let A be a nonzero ring and $n \geqslant 1$.

---

[1] In fact, the following three sets are in bijection with each other:
(1) $\{$submodules N' of M such that $M = N \oplus N'\}$;
(2) $\{p \in \mathrm{Hom}_A(M, N) \mid pi = \mathrm{id}_N\}$;
(3) $\{j \in \mathrm{Hom}_A(M/N, M) \mid qj = \mathrm{id}_{M/N}\}$.

(1) Show that if $v_1, \ldots, v_n$ generate A^n as an A-module, then they are linearly independent over A (hence form an A-basis of A^n). [2]

(2) Find an example to show that given $v_1, \ldots, v_n \in A^n$ which are linearly independent over A, they need *not* generate A^n as an A-module. [3]

8. This problem is about showing that the Baer–Specker group $\prod_{i=1}^{\infty} \mathbb{Z}$ is **not** free as a $\mathbb{Z}$-module.

(1) Show that for any $f \in \mathrm{Hom}_{\mathbb{Z}}\left(\prod_{i=1}^{\infty} \mathbb{Z}\,,\, \mathbb{Z}\right)$, there exists an $N > 0$ such that $f(e_n) = 0$ for all $n \geqslant N$. Here, e_n is the element in $\prod_{i=1}^{\infty} \mathbb{Z}$ whose sole nonzero entry is a 1 in the n-th position.

Hint: Denote $a_i = f(e_i) \in \mathbb{Z}$. Choose positive integers $n_1 < n_2 < \ldots$ such that for any $m \geqslant 1$, one has

$$\sum_{i=1}^{m} |a_i| 2^{n_i} \leqslant \frac{1}{2} 2^{n_{m+1}}.$$

Define

$$\mathbf{x} \coloneqq (2^{n_1}, 2^{n_2}, \ldots) \in \prod_{i=1}^{\infty} \mathbb{Z}.$$

Use $f(\mathbf{x})$ to show

$$f(0, \ldots, 0, 1, 2^{n_{m+2} - n_{m+1}}, 2^{n_{m+3} - n_{m+1}}, \ldots) = 0$$

for $m \gg 0$. Finally, show $f(e_{m+1}) = 0$ for $m \gg 0$.

(2) Show that if $f \in \mathrm{Hom}_{\mathbb{Z}}\left(\prod_{i=1}^{\infty} \mathbb{Z}\,,\, \mathbb{Z}\right)$ and f vanishes on the subgroup $\bigoplus_{i=1}^{\infty} \mathbb{Z}$, then $f \equiv 0$.

Hint: Consider elements of the form $\mathbf{x} = (2a_1, 4a_2, 8a_3, \ldots) = (2^i a_i)_{i \geqslant 1}$ with $a_i \in \mathbb{Z}$. Use $f|_{\bigoplus_{i=1}^{\infty} \mathbb{Z}} = 0$ to show $f(\mathbf{x}) = 0$. Similarly, $f(\mathbf{y}) = 0$ for all $\mathbf{y} = (3^i b_i)_{i \geqslant 1}$.

(3) Show that the following is an isomorphism of $\mathbb{Z}$-modules

$$\begin{aligned} \bigoplus_{i=1}^{\infty} \mathbb{Z} &\longrightarrow \mathrm{Hom}_{\mathbb{Z}}\left(\prod_{i=1}^{\infty} \mathbb{Z}\,,\, \mathbb{Z}\right); \\ (n_i)_{i \geqslant 1} &\longmapsto \left((m_i)_{i \geqslant 1} \mapsto \sum_{i=1}^{\infty} m_i n_i\right). \end{aligned}$$

Hint: Use (1) and (2).

(4) Show that $\prod_{i=1}^{\infty} \mathbb{Z}$ is not a free $\mathbb{Z}$-module.

Hint: If $\prod_{i=1}^{\infty} \mathbb{Z} \cong \mathbb{Z}^{(I)}$ for some set I, then $\mathrm{Hom}_{\mathbb{Z}}\left(\prod_{i=1}^{\infty} \mathbb{Z}\,,\, \mathbb{Z}\right) \cong \mathbb{Z}^I$. Compare the cardinalities of $\mathbb{Z}^{(I)}$, $\mathbb{Z}^I$, $\prod_{i=1}^{\infty} \mathbb{Z}$, and $\bigoplus_{i=1}^{\infty} \mathbb{Z}$.

[2] Equivalently, a surjective A-linear map $\varphi : A^n \to A^n$ must be injective (hence an isomorphism).

[3] Equivalently, an injective A-linear map $\varphi : A^n \to A^n$ need *not* be surjective.